

BDA 2 marks

1. Two Characteristics of Big Data with Suitable Examples

Volume refers to an enormous size of data, such as massive digital archives of scanned medical records or customer correspondence. **Velocity** describes the rapid accumulation speed as data dynamically flows in from networks, social media platforms, or real-time road cameras and traffic surveillance devices. (48 words)

2. Difference Between Descriptive Analytics and Diagnostic Analytics

Descriptive analytics examines historical data and past performance to understand what happened, utilizing tools like interactive data dashboards or standard sales reports. Conversely, **diagnostic analytics** drills deeper into data using discovery and correlation techniques to explain why a specific business problem or operational failure occurred. (49 words)

3. Support Processing of Large Datasets Using Hadoop Features

Hadoop's **distributed processing** coordinates parallel analysis across multiple inexpensive commodity machines, greatly enhancing overall processing speeds. This is supported by **data locality**, a feature that transmits processing queries directly to the specific servers holding the target data blocks rather than moving massive datasets across networks. (49 words)

4. Major Components of the Hadoop Ecosystem for Processing

The foundational components of the **Hadoop ecosystem** include **HDFS** for reliable, distributed data storage, and **MapReduce** or **Spark** for programmatic parallel and fast in-memory data processing. Additionally, **YARN** functions as the operating system, managing cluster resources and scheduling various data processing tasks. (49 words)

5. HDFS Commands for Uploading and Downloading Files

To upload files from the local operating system to the distributed HDFS store, you use the **-put** or **-copyFromLocal** commands. To download files from HDFS back to the local system, you utilize the **-get** or **-copyToLocal** commands. (47 words)

6. Concept of Big Data and its Major Characteristics

Big Data describes extremely complex, massive datasets that exceed traditional processing software capabilities. Its major characteristics are the 6 V's: **Volume** (e.g., medical archives), **Velocity** (e.g., social

media streams), **Variety** (different formats like XML or HTML), **Veracity** (messy data quality), **Value**, and **Variability**. (50 words)

7. The 3 V's of Big Data with Suitable Examples

The core 3 V's of Big Data are: **Volume**, representing massive scale such as archives of medical forms or customer correspondence; **Velocity**, indicating the high speed of accumulation from social media or mobile locations; and **Variety**, highlighting diverse formats including HTML, CSV, and XML. (50 words)

8. Utilizing HDFS to Store and Manage Large-Scale Data

To manage large-scale data, **HDFS** splits massive files into default 128MB data blocks and distributes them across cluster slave nodes. It maintains high fault tolerance by replicating each block, typically three times, across different hardware machines to ensure the data remains safe if hardware fails. (49 words)

9. Identifying the Data Node Component in HDFS

The HDFS **DataNode** is the slave daemon running on inexpensive commodity hardware in a distributed environment. Its main role is to store the actual block-size chunks of files and execute block-level read, write, replication, and deletion tasks based on instructions from the master NameNode. (50 words)

10. NoSQL Data Model for Key-Value Pairs

The **Key-value store** NoSQL model is suitable. It stores data as a collection of linked pairs consisting of a simple string key (which acts as a unique identifier) and an opaque value. The value can represent any data type, and examples include **Riak**, **Amazon S3 (Dynamo)**, and **Redis**. (50 words)

11. Explain the Major Characteristics of Big Data

The major characteristics of Big Data are defined by the 6 V's: **Volume** (enormous size), **Velocity** (high speed of accumulation), **Variety** (structured, semi-structured, and unstructured formats), **Veracity** (data inconsistency), **Value** (valuable insights), and **Variability** (how fast data structures change). (50 words)

12. Role of Big Data Technologies in Handling Large-Scale Data

Big Data technologies are specialized software utilities designed to analyze, process, and extract information from massive, complex datasets that traditional systems fail to handle. They support both real-time streams and batch processing, leveraging distributed frameworks like **Hadoop** for scalable storage and **Apache Spark** for rapid in-memory computation. (52 words)

13. File-Storage Management Using NameNode and DataNode

The master **NameNode** manages the HDFS namespace, storing metadata such as file names, permissions, and exact block locations. Meanwhile, slave **DataNodes** store the actual block-size chunks of files and execute block-level read, write, and replication operations on commodity hardware as directed by the NameNode. *(49 words)*

14. Apply Hadoop Pipes for Processing Using C++

Hadoop Pipes provides a native C++ interface, allowing developers to write high-performance mapper and reducer functions in C++. Input data is read from HDFS as key-value pairs and processed using custom C++ logic, enabling developers to leverage lower-level memory management and existing C++ codebases and libraries. *(52 words)*

15. Identify Replication in NoSQL Databases

Replication in NoSQL databases copies and stores data across multiple local or remote servers to ensure high reliability, data availability, and fault tolerance if nodes fail. Its main designs include **master-slave replication** (only master processes updates) and **peer-to-peer replication** (all nodes accept reads/writes as equals). *(49 words)*
